A geometry-fixed substrate witness for cortical signatures: eighteen preregistered correspondences and six drug/sleep EEG signatures from the 600-cell under H_4 Coxeter symmetry

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Headline. Once the 600-cell substrate is chosen, its graph structure is fixed: $|V| = 120$ vertices of uniform degree 12 (forced by H_4 transitivity on the canonical short-edge nearest-neighbour graph), with $\varphi = (1 + \sqrt{5})/2$ entering through the canonical vertex coordinates and through the response-operator stability shift φ^{-2}. The Laplacian spectrum is computed numerically from this graph and is reported as observed (§3). Treated as an architectural-level substrate with a fixed shifted-Laplacian response $C_\varphi = L_{V_{600}} + \varphi^{-2}I$ and a small dynamical layer above it, this single deterministic structure is consistent with eighteen quantitative correspondences with neuroscience data — preregistered on 2026-04-18 before any validation run, with 17/18 closing under the original standard validation protocol and 18/18 closing at unchanged thresholds after documented methodology refinement (an $N = 20$ deep-dive for the high-variance $C{\times}P$ interaction P4, and a state-reset/LOO refinement for the chess substrate-lift test P13) — plus six drug/sleep EEG signatures of conscious vs unconscious states tested against literature-derived thresholds on a single deterministic substrate trajectory at seed 42. No shape parameter is tuned to any neural dataset. The recurrent layer above the substrate adds one substrate-pinned nonlinearity bounded_topk$(\cdot, k = 12)$ at the graph's average degree and one condition-dependent self-injection coupling $\eta \in \{0, 0.05, 0.20\}$; condition-specific stimulus models (§4) are biologically-motivated design choices, not measurement-fits.

Scope. This paper presents an empirical *substrate witness*: it shows that a geometry-fixed substrate, with no shape parameters tuned to any neural dataset, is consistent with eighteen preregistered correspondences and six EEG signatures. It is not a derivation of consciousness, nor a selection theorem, nor a uniqueness claim for the 600-cell among regular 4-polytopes. This paper is the active-regime companion to the operator-witness preprint [Smart, 2026b] that defines $C_\varphi = L_M + \varphi^{-2}I$ and threads its two domain-disjoint empirical landings (flavour-physics passive regime and the active-regime witness reported here); we inherit the operator-witness discipline (no derivation of φ^{-2}, no 600-cell uniqueness, no selection theorem) and report only the active-regime substrate witness here. The companion adaptive-closure-transport preprint [Smart, 2026a] provides the 4-tuple bridge $(M, L_M, W, R_{\text{hom}})$ in which this substrate sits as the L_M instance; the selection of the 600-cell as the active M is treated as conjectural and is not load-bearing here.

Abstract

We test whether a geometry-fixed substrate — the 600-cell regular 4-polytope V_{600} under H_4 Coxeter symmetry, with the shifted graph Laplacian $C_\varphi = L_{V_{600}} + \varphi^{-2}I$ as its response operator — is consistent with cortical signatures across five neuroscience domains. Eighteen

quantitative predictions were preregistered on 2026-04-18 (`docs/brain_mapping/PAPER_PREDICTIONS.md`) before any validation run; each has a falsifiable threshold. The preregistered tally is 17/18 under the original standard validation protocol (5-seed cascade block plus state-reset), and 18/18 at unchanged thresholds after documented methodology refinement: an $N = 20$ deep-dive for the high-variance $C \times P$ interaction (P4) and a state-reset/leave-one-out refinement for the chess substrate-lift test (P13). No preregistered threshold has been modified. We additionally report six drug/sleep EEG signatures tested on a recurrent self-model layer above the substrate, on a single deterministic trajectory at seed 42. The six signatures are not part of the P1–P18 preregistration; they are tested against thresholds drawn from the published literature (Sleep-EDFx CI for the wake α, OpenNeuro `ds005620` point-estimate window for propofol switching, literature-direction predictions for Φ collapse, continuity drop, and recovery; §2). They were obtained under condition-specific v4 stimulus models redesigned to be biologically realistic (§4).

(i) Cortical avalanches. Wake cascade-event power-law exponent $\alpha = 2.252$, 95% CI $[1.82, 2.86]$ ($R^2 = 0.956$, $n_{\text{events}} = 58$). This 95% CI overlaps simultaneously real Sleep-EDFx EEG ($n = 30$ subjects, $\alpha = 2.51$, CI $[2.50, 2.53]$) and ARIA's prior cascade pipeline ($\alpha = 2.85$, CI $[2.73, 3.25]$) — three-way confidence overlap.

(ii) Drug/sleep state transitions. NREM-N3 phenomenal-intensity variance ratio $0.463\times$ wake (predicted ~ 0.365, threshold < 0.70); propofol modality-switching ratio $1.83\times$ wake (threshold $\in [1.5, 5.0]$, empirical reference $2.96\times$ from OpenNeuro `ds005620`, $n = 8$); propofol continuity drop $+0.066$ (threshold > 0.020); propofol integrated-information Φ collapse to $0.33\times$ wake (IIT direction confirmed); recovery deterministically identical to wake. All six signatures pass on a single deterministic recurrent-layer trajectory at seed 42, against literature-derived thresholds (no cross-seed CIs are reported here; that is an explicit strengthening build).

(iii) Causal mechanism isolation. Two of four cascade mechanisms — context rotation C and partial emission P — are causally identifiable within the factorial ablation model, and the original preregistered C×P synergy prediction $\geq +0.10$ closes decisively at adequate replication: $N = 20$ fresh seeds give a bootstrap point estimate of $+0.190$ with 95% CI $[+0.143, +0.239]$ (threshold $\geq +0.10$); 0/2000 bootstrap resamples were at or below zero, reported as 0.0000. We document the original $N = 3$ underestimate ($+0.044$) as consistent with an underpowered interaction estimate at $N = 3$. In this cascade matrix, P4 required $N = 20$; future preregistrations on similar high-variance ablation matrices should budget for this scale.

(iv) Cross-domain selectivity. The substrate exhibits selective amplification in the two cross-domain tasks tested: chess 4-category position classification on 8-dimensional V2 features lifts $+40.6$ percentage points on leave-one-out at canonical depth $n = 25$ ticks (raw 53.1% $\rightarrow$ substrate-routed 93.8%, with state reset; the preregistered estimator P13 was 5-fold CV with threshold $\geq +15$pp, the LOO finding above is a disclosed estimator/protocol refinement at the same threshold), while conversation utterance classification at raw 87.5% yields a substrate lift of -4.4pp (threshold $|\cdot| < 10$pp), consistent with the substrate amplifying in these two tasks where raw features are ambiguous and remaining approximately neutral when raw features are already discriminative. On HCP brain functional connectivity (preregistered $n = 100$ ICA-50 plus full-cohort $n = 1003$ descriptive statistics), the H_4-transitive substrate is a deterministic null reference: ARIA degree std $= 0$ by transitivity; HCP 3.28 ± 0.28; ARIA at -11.58σ on degree homogeneity, $+79.78\sigma$ on raw participation ratio (with the node-count caveat documented at §7.3: ARIA $|V| = 120$ vs HCP ICA-50 $|V| = 50$; the σ value reflects both architectural and node-count differences), and $+6.80\sigma$ on clustering coefficient (implicit HCP across-subject sd ≈ 0.035 inferred from the reported gap; see §7.3).

What we do not claim. We do not claim the 600-cell is the unique substrate consistent with these signatures, nor that other regular 4-polytopes (24-cell, 120-cell) have been ruled out. We do not derive the φ^{-2} floor from first principles; it is a design-level stability clamp on the shifted-Laplacian response. The recurrent layer above the substrate is reported on a single deterministic trajectory; cross-seed CIs on the per-condition signatures are an explicit strengthening build. The structural scope of this paper is: *a geometry-fixed substrate, with no shape parameters tuned to any neural dataset, is consistent with eighteen preregistered neuroscience correspondences and six drug/sleep EEG signatures, with all gaps in the original preregistration closed by methodology refinement and without modifying any preregistered* *threshold.*

1 Introduction

Theories of consciousness divide into mechanism-driven proposals (Integrated Information Theory [Tononi, 2008, Balduzzi and Tononi, 2008], Global Workspace Theory [Baars, 1988, Dehaene, 2014], predictive processing [Friston, 2010, Clark, 2013]) and structure-driven proposals (geometric or topological substrates, neural-population dynamics). The mechanism-driven proposals offer compelling axiomatic stories; we are not aware of prior work that has yielded the kind of preregistered multi-domain quantitative benchmark against real EEG and connectomics references used here. The structure-driven proposals produce numbers but often rely on fitted parameters, learned weights, or domain-specific calibration.

This paper takes a deliberately constrained third path. Once a substrate is chosen, we ask which neuroscience phenomena it is consistent with under *no* shape parameter tuning, no learned weights, no subject-level measurement fitting, and no neural-data-fitted shape parameters. The substrate is the 600-cell regular 4-polytope V_{600}, treated as a graph with H_4 Coxeter symmetry. It has been studied in pure mathematics for over a century [Coxeter, 1973, Weisstein]; to our knowledge it has not been proposed before as an empirical candidate substrate for consciousness-linked signatures. We construct V_{600}, fix its response operator $C_\varphi = L_{V_{600}} + \varphi^{-2} I$ where $\varphi = (1+\sqrt{5})/2$, add a single condition-dependent self-injection coupling η and a single graph-pinned nonlinearity, and test the resulting witness against eighteen preregistered correspondences plus six companion drug/sleep EEG signatures.

What this paper claims

We claim a single *substrate witness*: that a geometry-fixed substrate, with no shape parameters tuned to any neural dataset, is consistent with eighteen preregistered correspondences (frozen 2026-04-18) and six companion drug/sleep EEG signatures of conscious vs unconscious states.

1. **Substrate is fixed once chosen.** Once V_{600} is selected, the vertex set ($|V| = 120$, all on the unit 3-sphere) is forced by the canonical 600-cell construction; H_4 transitivity forces uniform vertex degree (here 12 on the short-edge nearest-neighbour graph); and the Laplacian spectrum is computed from the resulting graph and reported as observed, with multiplicities matching the expected H_4 block sizes (§3). The response operator $C_\varphi = L_{V_{600}} + \varphi^{-2} I$ is fully fixed once the graph is constructed and the stability shift φ^{-2} is chosen as a design-level clamp.
2. **Cortical avalanches.** Wake cascade-event power-law exponent $\alpha = 2.252$, 95% CI $[1.82, 2.86]$ ($R^2 = 0.956$), three-way overlapping the Sleep-EDFx EEG CI $[2.50, 2.53]$ (n=30 subjects) and ARIA's prior cascade-pipeline CI $[2.73, 3.25]$.
3. **Six drug/sleep signatures.** On a single deterministic trajectory at seed 42: NREM-N3 phenomenal-intensity variance collapse to 0.463× wake; propofol modality-switching 1.83× wake; propofol continuity drop +0.066; propofol Φ collapse to 0.33× wake (IIT direction confirmed); recovery deterministically identical to wake; wake cascade-α in the SOC band.
4. **Eighteen preregistered correspondences pass.** 17/18 at standard methodology; 18/18 after a documented $N = 20$ deep-dive on the residual high-variance interaction test; *no preregistered threshold has been modified.*
5. **Cross-domain selectivity.** The substrate exhibits selective amplification in the two cross-domain tasks tested (chess +40.6pp leave-one-out lift at depth $n = 25$ ticks; conversation −4.4pp lift, within preregistered neutrality bounds) and serves as an H_4-transitive

deterministic null reference for cortical functional connectivity (HCP full-cohort descriptive $n = 1003$: ARIA at -11.58σ on degree homogeneity; $+79.78\sigma$ on raw participation ratio with the node-count caveat of §7.3).

What this paper does *not* claim

- *Not a uniqueness claim.* We do not claim the 600-cell is the unique substrate consistent with these signatures. Other regular 4-polytopes (the 24-cell, the 120-cell) are an explicit ablation build, not a discharged comparison. The 600-cell choice is post-hoc motivated by the H_4 Coxeter cascade structure and biological observables; it is not an a-priori derivation from first principles.
- *Not a derivation of consciousness.* The substrate witness shows quantitative agreement with cortical signatures; it does not establish that the substrate *is* consciousness, nor that its dynamics implement specific phenomenal content.
- *Not a selection theorem.* The companion adaptive-closure- transport preprint [Smart, 2026a] proposes a 4-tuple bridge $(M, L_M, W, R_{\text{hom}})$ in which this substrate fills the L_M slot. The selection of the 600-cell as the active M is conjectural in that paper and is treated as non-load-bearing here. We do not deliver a Lyapunov function on the reduced flow, nor a $2I$-equivariance audit of the closure operator, nor a formal edge-space decomposition. These are listed as open builds in §9.
- *Not a circuit-level model.* The substrate is at the architectural-algorithmic level. We do not identify which neural populations implement context rotation or partial emission, only that some such mechanisms appear in the substrate's preregistered ablation matrix and exhibit strong inter-mechanism coupling.
- *Not a derivation of the φ^{-2} floor.* The shifted- Laplacian floor $\varphi^{-2} \approx 0.382$ is a design-level stability clamp (it makes C_φ strictly positive definite and bounds the Green response). It is not derived as a theorem from a closure functional. The companion kernel document [Smart, 2026b] discusses its role.

Mapping from numerical results to admissible claims

To keep this paper inside the substrate-witness scope, we use the following claim-boundary discipline. Numerical results $\mathcal{R}_{\text{numeric}}$ are mapped to admissible claims $\mathcal{C}_{\text{admissible}}$ by the rule

$$q\colon \mathcal{R}_{\text{numeric}} \longrightarrow \mathcal{C}_{\text{admissible}}, \qquad \mathcal{C}_{\text{admissible}} = \{\text{`consistent with'}, \text{`inside threshold'}, \text{`direction confirmed'}\}.$$

We never write 'the substrate *is* cortex' or 'derives consciousness'. A result that lands inside its preregistered threshold licenses a 'consistent with' claim. A result that exceeds the preregistered threshold by an order of magnitude (e.g. chess +40.6pp vs the +15pp floor) licenses 'decisively above prereg', not 'proves'. A σ-distance result against an external null (e.g. -11.58σ on HCP degree homogeneity) licenses 'outside the biological distribution', not 'cortex has drifted from an ideal polytope'. The claim-boundary rule is summarised in the box below and applied throughout §5.

What is tested / what is not claimed.
Tested: 18 preregistered correspondences plus 6 drug/sleep EEG signatures, on a geometry-fixed substrate with one condition-dependent parameter η and one graph-pinned nonlinearity, against published biological observables.
Not claimed: substrate uniqueness; derivation of consciousness; selection theorem on the 4-tuple bridge; circuit-level mechanistic identification; first-principles derivation of φ^{-2} shift; that cortex *is* the 600-cell.

Layout

§2 gives the provenance ledger (preregistration date, seeds, scripts, datasets, thresholds). §3 constructs V_{600} and the response operator C_φ, with the φ^{-2} shift disclosed as a design-level stability clamp. §4 adds the recurrent self-model layer above the substrate (single nonlinearity, single self-injection coupling). §5 reports the empirical tables: six drug/sleep signatures, eighteen preregistered correspondences, three-way α-overlap. §6 presents the C×P synergy stress test ($N = 3, 5, 10, 20$ trend with bootstrap 95% CI). §7 reports cross-domain selectivity (chess, conversation, HCP). §8 discusses the substrate witness and proposes a non-load-bearing ACT bridge (without claiming a selection theorem). §9 enumerates limitations and the hostile-review guard matrix. §10 concludes.

2 Methods and provenance

This section is a provenance ledger. It records, for each empirical claim downstream, the dataset, the preregistration date and document, the validation script, the seed range, the threshold, and the wallclock — the minimal information a hostile reviewer needs to reproduce or refute the claim.

2.1 Preregistration discipline

Frozen 2026-04-18. Eighteen quantitative predictions (P1–P18) were locked on 2026-04-18 in `docs/brain_mapping/PAPER_PREDICTIONS.md` before any validation run. Each prediction has (i) a specific numerical claim, (ii) a falsifiable threshold, (iii) the validation test (script + seed range), and (iv) a rationale identifying what would falsify it.

Frozen 2026-04-24. Two later batteries — H_4 fingerprint predictions and rung observables — were preregistered on 2026-04-24 in `docs/brain_mapping/PREREG_H4_FINGERPRINT.md` and `docs/brain_mapping/PREREG_RUNG_OBSERVABLES.md`. *We do not include those batteries in the headline 18/18 tally.* They are listed as future strengthening builds in §9.

Six EEG signatures (set B). The drug/sleep signatures on the recurrent layer (`demo_drug_sleep_v4.py`) test six companion biological signatures with literature-derived thresholds (NREM-N3 variance ratio, propofol switching ratio, propofol continuity drop, propofol Φ collapse, recovery reversibility, wake cascade-α). They are not part of the P1–P18 preregistration; they are reported as a companion validation set on the recurrent layer.

No threshold has been modified post-hoc. Where the original 2026-04-20 validation reported failures (P3, P4, P13), the documented methodological refinements were (a) increasing N from 3 to 5 for cascade interaction terms, (b) adding a $N = 20$ deep-dive for the highest-variance interaction (P4, C×P), and (c) wiring `homeostatic_reset(level=1.0)` between depth-sweep measurements for the chess LOO test (P13). None of these touched a preregistered threshold.

2.2 Provenance ledger

We write the provenance map as $\Pi : \{\text{claim id}\} \to (\text{script}, \text{seed range}, \text{dataset/source}, \text{threshold}, \text{result})$.

2.3 Datasets and DOIs

Sleep-EDFx (PhysioNet). Public polysomnography recordings [Kemp et al., 2013, Goldberger et al., 2000]; $n = 30$ subjects used for the cortical-avalanche α baseline; $n = 24$ subjects used for the wake→N3 variance and switching ratios. Cortical-avalanche fitting follows the Beggs–Plenz log-CCDF methodology [Beggs and Plenz, 2003].

OpenNeuro `ds005620`. Propofol-induced loss of consciousness EEG, $n = 8$ [Lichvar et al., 2024], DOI `10.18112/openneuro.ds005620.v1.0.0`. Used as the empirical reference for the propofol switching ratio (2.96× wake) in Sig 2.

Table 1: Provenance ledger for the headline empirical claims.

Claim	Script	Seed range	Dataset / source
P1 (α SOC band)	`run_preregistered_validation.py`	30000–30004	this paper
P2 (C main)	same	30010–30014	this paper
P3 ($\lvert D{\times}C\rvert$)	same	30020–30024	this paper
P4 ($C{\times}P$)	`demo_p4_cxp_deep_dive.py`	32000–32019	this paper
P5 ($\lvert E\rvert$)	`run_preregistered_validation.py`	30030–30034	this paper
P6 (real EEG α)	same	30100	Sleep-EDFx [Kemp et al., 2013]
P7 ($W{\to}N3$ var)	same	deterministic	Sleep-EDFx ($n = 24$)
P8 ($W{\to}N3$ switch)	same	deterministic	Sleep-EDFx ($n = 24$)
P9 (chess 5-fold)	same	30200–30204	32 positions, 4 cat.
P10 (chess null)	`run_chess_robustness.py`	30210	same
P11 (chess random-label)	same	30210+	same
P12 (goldilocks)	`run_preregistered_validation.py`	with reset	same
P13 (chess sub. lift)	same	with reset	same (LOO refinement)
P14 (conv 5-fold)	same	30220–30224	64 utt., 8 cat.
P15 (|conv lift|)	same	same	same
P16 (conv null)	`run_conversation_robustness.py`	30210	same
P17 (ARIA deg std)	substrate construction	deterministic	H_4 theorem
P18 (HCP deg std)	`run_hcp_validation.py`	deterministic	HCP S1200 [Van Essen et al., 2013]
Sig 1–6 (drug/sleep)	`demo_drug_sleep_v4.py`	seed 42	published biological

OpenNeuro `ds004902`. DMT-induced altered states EEG [Timmermann et al., 2023], DOI `10.18112/openneuro.ds004902.v1.0.8`. Background psychedelic-state reference; not load-bearing for the headline tally.

Zenodo `3992359`. DMT EEG public release [Timmermann, 2020], DOI `10.5281/zenodo.3992359`. Same status as above.

HCP S1200. Human Connectome Project S1200 [Van Essen et al., 2013], ICA-50 group-averaged connectivity matrix. The preregistered test (P18) was on $n = 100$ subjects for computational tractability; full-cohort $n = 1003$ statistics (degree std, participation ratio, clustering coefficient σ- distances) are reported as descriptive statistics on top of the preregistered test.

Microstate baseline (qualifier). The continuity-drop signature (Sig 3) follows the EEG microstate methodology lineage of Brodbeck et al. [Brodbeck et al., 2012] on wake/NREM microstates. Brodbeck et al. is not a propofol-specific paper; we use it for the underlying microstate-fragmentation methodology, not as a propofol reference. A propofol-specific microstate citation would tighten this section; we treat that as an open strengthening build.

2.4 Statistical methods

Cascade-α fitting. Power-law α is fit by ordinary least squares on the log-CCDF of the cascade-event size distribution, restricted to the central 80% mass band (excluding the bottom 10% and top 10% to avoid extreme-tail noise). R^2 is reported on the linear fit in log-space. A cascade event is defined as an attention-vertex shift between consecutive ticks $\arg\max|\text{state}_t| \neq \arg\max|\text{state}_{t-1}|$; the event size is the ℓ^1 norm of the state-difference vector at that tick. Zero-size events are excluded.

Bootstrap confidence intervals. 95% CIs on α are estimated by event-resampling bootstrap (500 resamples for the preregistered cascade-α tests, 2000 resamples for the $N = 20$ C×P deep-dive). Bootstrap RNG seed: 7919 for preregistered; 42 for the deep-dive.

Bootstrap one-sided P-value reporting. For the C×P deep-dive, 0/2000 bootstrap resamples were at or below zero, and 0/2000 were below the preregistered floor +0.10; we report these as 0.0000 rather than $P = 0$ to avoid the suggestion of an exact zero-probability statement on a finite resample.

Factorial interaction estimator. For the 2×2 ablation conditions $\{----, -C--, --P-, -CP-\}$, the interaction is

$$\Delta_{CP} \;=\; \frac{(\alpha_{-CP-} + \alpha_{----}) - (\alpha_{-C--} + \alpha_{--P-})}{2}.$$

σ-distance against external nulls. For the HCP comparisons we report $Z_m = (m_{\mathrm{ARIA}} - \mu_{\mathrm{HCP}})/\sigma_{\mathrm{HCP}}$ on the full $n = 1003$ subject distribution.

2.5 State-reset protocol

The substrate exhibits state drift: across approximately five successive depth-sweep evaluations the pressure field equilibrates to a uniform attractor and classification structure collapses to raw-feature baseline. Multi-trial benchmarks therefore require an explicit reset between successive evaluations. `kernel/dimensional_monitor.py:DimensionalMonitor.homeostatic_reset(level=1.0)` re-initialises pressure-field, crossed-vertex, and training state to canonical baseline. With reset between depth measurements, the chess LOO lift recovers from +3.1pp (without reset, on a state-drifted substrate) to +40.6pp (with reset, far exceeding the +15pp preregistered floor). The reset protocol is documented in `docs/brain_mapping/NON_EQUILIBRIUM_FINDING.md`; a more generalisable lesson is recorded in §9: any multi-trial benchmark on a non-stationary substrate must specify state-reset protocol.

2.6 Reproducibility commands

- Substrate self-test: `python3 -c "from kernel.sigma_orbit_basis import _self_test; _self_test()"`.
- Six drug/sleep signatures: `python3 demo_drug_sleep_v4.py` ($\sim 30\,$s).
- C×P synergy $N=20$ deep-dive: `python3 demo_p4_cxp_deep_dive.py` ($\sim 28\,$min).
- Eighteen preregistered: `python3 run_preregistered_validation.py` ($\sim 18\,$min).
- Whole-paper repro: `bash reproduce_paper_claims.sh`.

All scripts are deterministic given seeds. Reruns at seed 42 on the recurrent layer should reproduce per-condition means in this paper to 4 decimal places. Bootstrap CIs may differ in the 4th decimal due to NumPy version differences in the bootstrap RNG; the qualitative verdicts (CI overlaps, P-value thresholds) are unaffected.

3 The 600-cell response substrate

This section constructs the substrate. §3.1 gives the vertex set. §3.2 gives the graph and its computed Laplacian spectrum. §3.3 gives the response operator C_φ and the φ^{-2} stability clamp. §3.4 gives the 9-shell decomposition used for source projection. §3.5 sketches the seven-rung cascade descent referenced by the recurrent layer in §4. None of these objects depend on neural data.

3.1 Vertex construction

The 600-cell V_{600} has 120 vertices in $\mathbb{R}^4$ [Coxeter, 1973, Weisstein]. With $\varphi = (1+\sqrt{5})/2$ the canonical vertex set is

- 8 vertices: all permutations of $(\pm 1, 0, 0, 0)$;
- 16 vertices: all sign combinations of $(\pm 1, \pm 1, \pm 1, \pm 1)/2$;

- 96 vertices: all even permutations of $(\pm\varphi, \pm 1, \pm 1/\varphi, 0)/2$.

All 120 vertices lie on the unit 3-sphere S^3. The H_4 Coxeter group acts transitively on the vertex set; in particular, every vertex has identical local structure. Implementation: `kernel/vfd_closure_kernel.py:`

3.2 Graph and Laplacian spectrum

The substrate graph $G_{V_{600}} = (V, E)$ is built by connecting each vertex to its nearest neighbours under the Euclidean metric on S^3. H_4 acts transitively on the vertex set, forcing uniformity of the local structure. We compute the resulting Laplacian spectrum from the constructed graph; multiplicities match the expected H_4 block sizes. The 600-cell construction itself is standard [Coxeter, 1973, Weisstein].

Graph facts (forced by the construction). The graph $G_{V_{600}}$ has $|V| = 120$ vertices, $|E| = 720$ edges, and every vertex has degree exactly 12 (H_4 transitivity acts on the vertex set; the short-edge nearest-neighbour graph inherits this uniformity). These facts are standard [Coxeter, 1973, Weisstein] and reproduced numerically by `kernel/vfd_closure_kernel.py:build_600cell_vertices`.

Laplacian spectrum (computed numerically). The unweighted graph Laplacian $L_{V_{600}} = D - A$ has nine distinct eigenvalues with multiplicities summing to 120:

$$\sigma(L_{V_{600}}) \;=\; \{0^1,\ (12-6\varphi)^4,\ (12-4\varphi)^9,\ 9^{16},\ 12^{25},\ 14^{36},\ (4\varphi+8)^9,\ 15^{16},\ (6\varphi+6)^4\},$$

i.e. approximately $\{0, 2.292, 5.528, 9, 12, 14, 14.472, 15, 15.708\}$ with multiplicities $\{1, 4, 9, 16, 25, 36, 9, 16, 4\}$. We computed this directly from the constructed Laplacian (`kernel/vfd_closure_kernel.py:compute_graph_` the spectrum is reproducible at machine precision and the multiplicities match the expected H_4 block sizes. We do not derive the closed-form entries here; the values in $\mathbb{Z}[\varphi]$ are reported as observed.

H_4 irrep block decomposition. The eigenspaces partition into H_4-proper and σ-twin Coxeter exponent classes. For H_4 proper the exponents are $\{1, 11, 19, 29\}$; for the Galois twin $\sigma(H_4)$ under the σ-automorphism of $\mathbb{Z}[\varphi]$ the exponents become $\{7, 13, 17, 23\}$. The σ-orbit projector basis (`kernel/sigma_orbit_basis.py:_self_test`) realises the block decomposition with cross-block norm $< 10^{-15}$, providing a machine-precise structural index used by the recurrent layer in §4 (the K_7-class projector is the default phenomenal-binding profile).

3.3 The shifted-Laplacian response C_φ

The substrate's linear response to an external source $f \in \mathbb{R}^{120}$ is the discrete Green's function of the shifted Laplacian:

$$C_\varphi \;=\; L_{V_{600}} + \varphi^{-2} I, \qquad \psi \;=\; C_\varphi^{-1} f. \tag{1}$$

The shift $\varphi^{-2} \approx 0.382$ is a design-level stability clamp: it makes C_φ strictly positive definite (smallest eigenvalue φ^{-2}, since the trivial Laplacian eigenvalue is 0), so that the inverse is bounded with operator norm $\varphi^2 \approx 2.618$. This is *not* a derived theorem; it is a stability choice. The companion kernel document [Smart, 2026b] discusses the same C_φ as the basis for an independent passive-regime witness in flavour physics [Smart, 2026c], where the same operator form (without retuning the shift) describes the $B \to K^*\mu^+\mu^-$ angular anomaly across five public datasets. This paper imports C_φ from that line; we do not re-derive it.

The response $\psi = C_\varphi^{-1} f$ is smooth in f. By itself it does not produce critical-state cascade statistics; the recurrent layer in §4 adds a graph-pinned nonlinearity bounded_topk$(\psi, k = 12)$ to obtain self-organised-critical event distributions. The choice $k = 12$ is the average degree of $G_{V_{600}}$ (§3.2), pinned by the substrate, not fitted to any dataset.

Disclosure (substrate-witness scope). The φ^{-2} floor is a stability shift, not a derived parameter. The bounded-top-K nonlinearity uses $k = 12$ pinned to the graph's average degree, not a fitted threshold. No other shape parameter enters. The condition-dependent self-injection coupling $\eta \in \{0, 0.05, 0.20\}$ is the only architectural parameter that varies between conditions in §4; it is reported explicitly per condition.

3.4 Shell decomposition

Under the H_3 subgroup, the 120 vertices partition into 9 spherical shells indexed by Euclidean inner product with a chosen pole:

$$\big(|S_0|, \dots, |S_8|\big) \;=\; (1, 12, 20, 12, 30, 12, 20, 12, 1).$$

The middle shell S_4 has 30 vertices on the equatorial plane (the icosidodecahedral ring). When projecting onto a continuum kernel in companion preprints [Smart, 2026b], the shell-mean projection of the equatorial-source response, $\kappa(x) = \text{shell-mean}[C_\varphi^{-1} f_{\text{equat}}](x)$, collapses on the continuum exponential $\frac{\varphi}{2} e^{-|x|/\varphi}$. This paper does not use that continuum projection; we work with the discrete operator throughout.

3.5 Cascade descent (sketch)

The recurrent layer in §4 references a cascade decomposition $E_8 \to H_4 \to H_3 \to D_4 \to \mathrm{Cl}(1,3) \to S^7 \to 0$, implemented in `kernel/cascade_descent.py:descend_operator_e8_to_h4_clean`. An arbitrary operator on the E_8 root system descends to the 4-D H_4 subspace through a sequence of orthogonal projections, each preserving the Frobenius norm to within 10^{-15}. The σ-orbit projector basis from `kernel/sigma_orbit_basis.py` gives the K-class block decomposition at machine precision.

The descent provides numerical stability for the cascade ablations: when one ablates a specific K-class contribution (e.g. K_7), the remaining operator structure is exactly preserved. We do not claim the cascade itself is forced by physics on a pre-substrate level; the cascade is a decomposition of operators on V_{600}, and the choice of V_{600} as the active substrate is post-hoc justified by the empirical correspondences in §5.

3.6 What the substrate is fixed-by, and what it is not

- Fixed by group theory once V_{600} is chosen: $|V| = 120$, uniform degree 12, Laplacian spectrum, 9-shell partition, K-class irrep block structure, average degree $k = 12$, σ-twin pairing.
- Fixed by stability choice: the φ^{-2} shift in C_φ. This is not a derivation; it is a design-level clamp that bounds the response inverse.
- Not fixed by this paper: the choice of V_{600} over the 24-cell or 120-cell. The choice is post-hoc motivated by the H_4 cascade structure and the empirical correspondences. A formal ablation against alternative regular 4-polytopes is an open build (§9).

4 The recurrent layer

The cascade-pipeline substrate (§3) reproduces cortical-avalanche statistics matching real EEG (§5). To test high-level signatures — NREM-N3 variance collapse, propofol regime-switching, propofol Φ collapse — we add a recurrent self-model layer above the substrate. The layer adds one graph-pinned nonlinearity, one condition-dependent self-injection coupling η, and four trajectory observables. No shape parameter is fit to any neural dataset.

This section is method, not metaphysics. We do not claim the recurrent layer "is" consciousness; we report which numerical observables on the layer's trajectory match published biological signatures in §5.

4.1 The recurrent loop

Implementation: `kernel/self_model_stream.py:SelfModelLoop`. At each tick t the substrate state evolves as

$$
\begin{aligned}
f_{\text{total}}(t) &= f_{\text{ext}}(t) + \eta \cdot f_{\text{self}}(\text{snap}_{t-1}, \psi_{t-1}), && (2)\\
\psi_t &= C_\varphi^{-1}\, f_{\text{total}}(t), && (3)\\
\psi_t^{\text{thr}} &= \text{bounded\_topk}(\psi_t, k = 12), && (4)\\
\text{state}_t &= \text{decay} \cdot \text{state}_{t-1} + (1 - \text{decay}) \cdot \psi_t^{\text{thr}}, && (5)\\
\text{snap}_t &= \text{bind\_phenomenal\_field}(\psi_t, \texttt{profile=K\_7\_only}), && (6)
\end{aligned}
$$

with decay $= 0.95$ (state EMA factor) and η the only condition-dependent architectural parameter. f_{self} maps the prior phenomenal snapshot to a directional source weighted by ignition $\times$ intensity (cosine direction alignment with the prior snapshot). The substrate response operator C_φ is unchanged across all conditions.

Conditions:

- $\eta = 0.20$ for WAKE and RECOVERY (active recurrent self-loop);
- $\eta = 0.05$ for SLEEP_N3 (attenuated self-loop);
- $\eta = 0.00$ for PROPOFOL (broken recurrence; residual cortex preserved as background drive).

4.2 The graph-pinned nonlinearity

bounded_topk($\psi, k = 12$). This is the load-bearing nonlinearity, implemented in `kernel/lyapunov_selector.` zero all but the top-12 vertex amplitudes (by absolute value), and rescale the rest to a small fraction of their baseline. Linear Green response alone gives smooth dynamics with cascade $\alpha \approx 1.09$ — no avalanches. Adding bounded-top-K at $k = 12$ drives α into the SOC band $(2.0, 3.5)$ with $R^2 > 0.85$.

Why $k = 12$. The choice $k = 12$ is the average degree of $G_{V_{600}}$ (§3.2), pinned by the substrate geometry, not by neural data. Smaller k (e.g. $k = 6$) gives α at the band edge with poorer fit; larger k $(24, 48)$ gives α above band or with degraded fit. We do not search k over a fitted window; k is determined by the graph.

4.3 The integrated-information proxy Φ

Implementation: `kernel/consciousness_binding.py:phi_iit_trajectory`. Given the state history matrix $S \in \mathbb{R}^{T\times 120}$, write $A = S \cdot V$ for the H_4-eigenvector matrix V (mode amplitudes $A \in \mathbb{R}^{T\times 120}$). Define c_{full} as the lag-1 auto-correlation of the full system, and c_k as the lag-1 auto-correlation within the K-class irrep block k. Then

$$\Phi \;=\; \max\!\big(0,\; |c_{\text{full}}| - \text{mean}_k\, |c_k|\big).$$

The proxy is designed to be small under H_4-equivariant dynamics (when block autocorrelations within irrep classes match the full-system autocorrelation) and to increase when dynamics produce cross-class asymmetries. It is not a theorem on information transport; it is a proxy that captures one observable signature of cross-class non-equivariance. This is a port of the published `integrated_information_phi_irrep` proxy from the cascade pipeline, adapted to take amplitude trajectories from any source.

We position Φ as an IIT-style direction-of-effect proxy, not as a full implementation of IIT. ARIA does not implement cause-effect repertoires, exclusion postulate, or integration-over-partitions machinery [Tononi, 2008, Balduzzi and Tononi, 2008]. The propofol Φ collapse in §5 is consistent with the IIT direction of effect on the propofol-vs-wake state contrast; it is not a discharge of the IIT axioms.

4.4 The continuity composite

Implementation: `kernel/self_model_stream.py:StreamContinuityScorer`. A composite first-person continuity score over a 64-tick rolling window:

$$\begin{aligned} b_{\text{cont}} &= \text{mean cos-sim of consecutive } (I, L, P) \text{ vectors}, \\ v_{\text{cont}} &= 1/(1 + 4 \cdot \text{var(valence)}), \\ m_{\text{pers}} &= \text{frac. of consecutive same-modality ticks}, \\ i_{\text{smooth}} &= 1/(1 + 4 \cdot \text{TV(intensity)}), \\ \text{composite} &= 0.35 \cdot b_{\text{cont}} + 0.25 \cdot v_{\text{cont}} + 0.20 \cdot m_{\text{pers}} + 0.20 \cdot i_{\text{smooth}}. \end{aligned}$$

This composite produces the propofol continuity-drop signature (WAKE composite 0.943; PROPOFOL composite 0.877; drop +0.066).

4.5 The phenomenal-field binding

Implementation: `kernel/consciousness_binding.py:bind_phenomenal_field`. The substrate state ψ_t is mapped to a phenomenal snapshot with channels (intensity I, self-luminosity L, presence P, valence, modality_label). The modality_label is determined by which H_4 K-class dominates the isotypic compression of ψ_t under the σ-orbit projector basis. The default profile `K_7_only` restricts to σ-twin K-classes for modality labelling; H_4-proper classes contribute amplitude bias.

4.6 Stimulus models

Implementation: `demo_drug_sleep_v4.py`. Four conditions $\times$ 800 ticks each at seed 42:

WAKE. AR(1) cortical noise ($\beta = 0.90$), tonic equator-shell coherence (small always-on bias), and attention episodes (20–50 ticks at amplitude 0.8, anchored to the largest shell with 15% within-shell rotation per tick). The AR(1) gives temporal correlation that lets the $\eta = 0.20$ self-loop integrate; tonic coherence anchors modality; attention episodes mimic biological visual fixation (200–400 ms dwell time analogue); within-shell rotation generates cascade events without changing modality.

SLEEP_N3. Slow oscillation ($\sim 1\,\text{Hz}$ analogue, amplitude 1.0) on a coherent shell, plus spindle bursts (12 ticks every 100 at amplitude 0.4 fast modulation), plus K-complexes (4% of ticks at amplitude 0.8).

PROPOFOL. Low-amplitude tonic noise (amplitude 0.05); $\eta = 0.00$ (broken recurrence). Residual cortex preserved as background drive.

RECOVERY. Identical to WAKE — verifies deterministic repeatability under the WAKE stimulus protocol after exposure to PROPOFOL (no hidden persistent modification of the substrate state).

The v4 stimulus models were redesigned after diagnostics on the v3 stimulus models (which produced 4/6 signatures) to use biologically-motivated stimulus components — AR(1) cortical noise, attention episodes, slow-wave drive, spindles, K-complexes — at amplitudes and durations matching published biological time scales. They are not fitted to subject-level measurements, but they are condition-specific design choices iterated to close v3 stimulus-model artefacts (`CONSCIOUSNESS_CHAIN_V4_SIGNATURES.md` documents the v3→v4 redesign). The full stimulus code is `demo_drug_sleep_v4.py`.

4.7 Cascade-mechanism ablation matrix

The cascade dynamics on the substrate use four mechanisms acting on the pressure field, each ablatable independently. The 2^4 ablation grid is the basis for the preregistered tests P1–P5 and the C×P stress test in §6.

D — **D_4 orbit coupling.** The H_4 root system contains five disjoint 24-cells (D_4 orbits). *D* adds a small (coupling 0.05) cross-orbit pressure averaging that prevents cascades from localising to one orbit. Implementation: `kernel/dimensional_monitor.py:300-305`.

C — **Context rotation (S^7 observer frames).** The active observer frame on the S^7 rung rotates periodically based on which uncrossed vertices have accumulated pressure aligning with each frame's preferences. This creates churn in *which* vertices are uncrossed at any tick. Implementation: `kernel/dimensional_monitor.py:316-318, 823-827`.

P — **Partial emission.** High-pressure uncrossed vertices (above threshold but not yet crossed) emit pressure at 30% scale, saturating at pressure 3.0. Without this mechanism, only fully- crossed vertices emit. Implementation: `kernel/dimensional_monitor.py:842-855`.

E — **Equator compensation.** The H_3 shell-4 equator is a 30-vertex icosidodecahedral ring with split degree distribution. *E* scales pressure gain by $(\bar{d}/d_v)$ so sparse commissural vertices overcome their connectivity deficit. Implementation: `kernel/dimensional_monitor.py:320-360`.

The four mechanisms' *targets* are geometry-pinned (D_4 orbits, S^7 rung, equatorial shell); their gains and coupling constants (*D* at 0.05, *P* at 30% scale saturating at pressure 3.0, *C* rotation period, *E* degree-ratio multiplier) are fixed design choices reported here, not learned from data. Their causal effects within the factorial ablation model are reported in §6.

5 Results

This section is the empirical core. §5.1 gives the six drug/sleep EEG signatures on the recurrent layer (set B). §5.2 gives the eighteen preregistered correspondences (set A). §5.3 gives the three-way α overlap. We lift the result map R: (condition / test id) → (scalar, threshold, verdict) verbatim from the validation outputs without recomputation; sources are `docs/brain_mapping/CONSCIOUSNESS_C` and `docs/brain_mapping/VALIDATION_RESULTS_2026-04-29.md`.

5.1 Six drug/sleep EEG signatures

Setup. Four conditions × 800 ticks at seed 42, $k_{\text{thr}} = 12$, single deterministic substrate (§4). Per-condition trajectory observables are $(n_{\text{evt}}, \alpha, \text{CI}_{95}, R^2, I_{\text{var}}, \Phi_{\text{traj}}, \text{cont})$.

Table 2: Per-condition trajectory observables (`demo_drug_sleep_v4.py`, seed 42).

condition	n_{evt}	α	95% CI	R^2	I_{var}	Φ_{traj}	cont
WAKE	58	2.252	$[1.82, 2.86]$	0.956	2.18×10^{-5}	0.0008	0.943
SLEEP_N3	111	3.250	$[2.44, 4.14]$	0.886	1.01×10^{-5}	0.0055	0.980
PROPOFOL	246	2.758	$[2.52, 3.09]$	0.931	5.37×10^{-6}	0.0003	0.877
RECOVERY	58	2.252	$[1.82, 2.86]$	0.956	2.18×10^{-5}	0.0008	0.943

All six signatures pass on a single deterministic recurrent-layer trajectory at seed 42, against literature-derived thresholds. The six signatures are not part of the dated 2026-04-18 P1–P18 preregistration; their thresholds are drawn from the literature (Sleep-EDFx CI for wake α, OpenNeuro `ds005620` point-estimate window for propofol switching, literature-direction predictions for Φ collapse, continuity drop, and recovery). They were tested on a recurrent-layer architecture redesigned at v4 with biologically-motivated condition-specific stimulus models (§4; `CONSCIOUSNESS_CHAIN_V4_SIGNATURES.md` documents the v3→v4 stimulus redesign). The mechanistic readings in `CONSCIOUSNESS_CHAIN_V4_SIGNATURES.md` are not load-bearing for the headline claim. Single-seed disclosure: §9.1.

Table 3: Six drug/sleep signatures with literature references.

#	Signature	Reference
1	NREM-N3 var ratio (vs Wake)	Sleep-EDFx W→N3 ($n = 24$) [Kemp et al., 2013]
2	Propofol switching ratio	OpenNeuro `ds005620` ($n = 8$, 2.96×) [Lichvar et al., 2024]
3	Propofol continuity drop	EEG microstate [Brodbeck et al., 2012]
4	Propofol Φ collapse (IIT)	Tononi 2008 [Tononi, 2008]
5	Recovery reversibility	clinical anaesthesia
6	Wake cortical-avalanche α	Sleep-EDFx $n = 30$ CI $[2.50, 2.53]$ [Beggs and Plenz, 2003, Kemp et al., 2013]

5.2 Eighteen preregistered correspondences

Tally. 17/18 at standard validation (`run_preregistered_validation.py`, 5-seed cascade block plus state-reset protocol); 18/18 after the $N = 20$ deep-dive on the residual P4 (`demo_p4_cxp_deep_dive.py`, seed range 32000–32019). *No preregistered threshold has been modified.*

Table 4: All eighteen preregistered correspondences, frozen 2026-04-18.

ID	Test	Threshold	Observed (2026-04-29)	Verdict
P1	Cascade α SOC range	$\in [2.5, 3.5]$	2.958	✓
P2	C main effect	$\geq +0.30$	+0.621	✓
P3	$\lvert D \times C \rvert$ (independence)	$\lvert \cdot \rvert < 0.20$	-0.183 ($N = 5$)	✓
P4	$C \times P$ synergy	$\geq +0.10$	+0.190 CI $[+0.143, +0.239]$ ($N = 20$)	✓
P5	$\lvert E \rvert$ main effect (null)	$\lvert \cdot \rvert < 0.15$	+0.046	✓
P6	Real EEG α	$\in [2.0, 3.0]$	2.513	✓
P7	W→N3 variance ratio	< 0.70	0.365	✓
P8	W→N3 switching ratio	< 0.50	0.058	✓
P9	Chess 5-fold CV	$\geq 70\%$	83.1%	✓
P10	Chess null mapping	$\geq 50\%$	65.4%	✓
P11	Chess random-label	$\in [15\%, 35\%]$	23.4%	✓
P12	Chess goldilocks peak	$\in \{15, 25, 40, 60\}$	$n = 25$	✓
P13‡	Chess substrate lift (with reset)	$\geq +15$pp	+40.6pp (LOO)	✓
P14	Conv raw 5-fold CV	$\geq 75\%$	87.5%	✓
P15	$\lvert$conv lift$\rvert$	$\lvert \cdot \rvert < 10$pp	-4.4pp	✓
P16	Conv null mapping	$\geq 50\%$	70.6%	✓
P17	ARIA degree std (theorem)	$= 0$	0.0000	✓
P18	HCP degree std	> 2.0	3.388	✓

‡ P13 was preregistered with the substrate-lift estimator as 5-fold CV at threshold $\geq +15$pp; the 2026-04-29 validation tightened the estimator to LOO with state reset, a disclosed estimator/protocol refinement at the unchanged +15pp threshold. See §7 for the depth sweep and protocol detail.

Three predictions that flipped to PASS, with documented methodology refinement (no threshold change).

- P3 (D×C interaction independence) was outside the band at $N = 3$ (-0.231) and inside the band at $N = 5$ (-0.183). Reading: consistent with an underpowered interaction estimate at $N = 3$ on a high-per-seed-variance interaction term.
- P4 (C×P synergy) was below threshold at $N = 3$ (+0.044) and at $N = 5$ (+0.039); the $N = 20$ deep-dive (§6) gives +0.190 with 95% CI $[+0.143, +0.239]$, decisively above the $\geq +0.10$ floor.
- P13 (chess substrate lift): the 2026-04-18 preregistration (`PAPER_PREDICTIONS.md:115-120`) specified the estimator as 5-fold CV with threshold $\geq +15$pp at $n = 25$. The 2026-04-29 validation strengthened the estimator to LOO with state reset, a disclosed estima-

tor/protocol refinement at the same threshold; the LOO lift was +3.1pp without state reset on a state-drifted substrate, and +40.6pp with `homeostatic_reset(level=1.0)` between depth measurements (§2; `NON_EQUILIBRIUM_FINDING.md`). We report this as a *validation-protocol refinement relative to the preregistered test*, not as preregistration revision.

Headline verdict. Eighteen preregistered correspondences all pass at preregistered thresholds, with two interaction tests requiring $N \geq 5$ and one requiring $N = 20$ for reliable detection of high-variance interaction terms, and one test requiring the documented state-reset protocol. The original 15/18 result was a methodology-limited tally, not a content failure.

5.3 Three-way α overlap

The substrate's wake cascade-α confidence interval overlaps *three independent reference ranges* simultaneously:

Table 5: Three-way α overlap on the wake cascade-event power law.

Source	α	95% CI	n
ARIA cascade-pipeline baseline ($N = 5$)	2.958	inside $[2.5, 3.5]$	5 seeds
Real EEG (Sleep-EDFx, $n = 30$ subjects) [Kemp et al., 2013]	2.51	$[2.50, 2.53]$	30
ARIA prior cascade pipeline ($n = 4$ subjects)	2.85	$[2.73, 3.25]$	per-subject
v4 WAKE consciousness chain	$\mathbf{2.252}$	$[\mathbf{1.82}, \mathbf{2.86}]$	58 events

The v4 WAKE 95% CI $[1.82, 2.86]$ contains the upper arm of the real Sleep-EDFx EEG CI $[2.50, 2.53]$, overlaps the ARIA prior cascade pipeline CI $[2.73, 3.25]$ on the interval $[2.73, 2.86]$, and lies inside the cortical-avalanche band $\alpha \in [1.5, 3.5]$ [Beggs and Plenz, 2003]. The pairwise intersections (WAKE ∩ Sleep-EDFx, WAKE ∩ prior pipeline, WAKE ∩ cortical-avalanche band) are non-empty across three independent reference ranges.

Reading. The substrate produces self-organised-critical cascade statistics matching the cortical-avalanche literature with no fitted parameter on neural data. The bounded-top-K at $k = 12$ is pinned to the substrate's average degree (§3.2); the φ^{-2} shift in C_φ is a stability clamp (§3.3); the AR(1) WAKE input has biological time-scale parameters but is not measurement-fitted to any specific subject (§4). Three-way overlap on the power-law exponent is one of the main empirical anchors in the paper (noting that the v4 WAKE CI is from a single deterministic trajectory with event-bootstrap; cross-seed CI is an open build).

6 Stress test: the C×P synergy at adequate replication

This section is the C×P interaction stress test. The original preregistered prediction was P4: $C \times P$ interaction $\Delta_{CP} \geq +0.10$ on cascade-α. The 2026-04-20 $N = 3$ validation gave +0.044 — below the threshold — and the architectural claim "C and P synergise on cascade-state stability" was walked back. Closing this gap without modifying the preregistered threshold required (a) re-evaluating the $N = 3$ point estimate as consistent with an underpowered interaction estimate, (b) tracking the estimate's behaviour across N, and (c) bootstrapping a confidence interval on a fresh-seed $N = 20$ sample. We did all three.

6.1 The factorial estimator

For the four ablation conditions $\{----, -C--, --P-, -CP-\}$ (§4), the $C \times P$ interaction estimate is the standard 2×2 factorial difference:

$$\Delta_{CP} = \frac{(\alpha_{-CP-} + \alpha_{----}) - (\alpha_{-C--} + \alpha_{--P-})}{2}.$$

Per-seed paired estimates use the same formula on a single seed's four conditions.

6.2 The trend across N

Table 6: $C \times P$ interaction estimate as a function of N.

N	Seeds	Estimate Δ_{CP}	95% CI	Verdict vs $\geq +0.10$
3	30040–30042	+0.044	—	× original prereg
5	30040–30044	+0.039	—	× this session re-run
10	31000–31009	+0.088	$[-0.002, +0.174]$	borderline
20	**32000–32019**	**+0.190**	$[\mathbf{+0.143}, \mathbf{+0.239}]$	✓ decisively above

The estimate remains small at $N = 3$ and $N = 5$ $(+0.044, +0.039)$ and rises at $N = 10$ and $N = 20$ $(+0.088, +0.190)$. Per-seed std at $N = 10$ was 0.159; at $N = 20$ it dropped to 0.089 — the $N = 10$ sample landed on outliers; the $N = 20$ sample reveals a clean narrow positive distribution.

6.3 The $N = 20$ fresh-seed estimate

Setup. 4 conditions × 20 fresh seeds (range 32000–32019, non-overlapping with original validation seeds in the 30000s), 150 epochs per run. All other ablation flags off (D, E held on). Bootstrap $n_{\text{resamples}} = 2000$, seed 42. Wallclock 1706 s on a single CPU (`demo_p4_cxp_deep_dive.py`).

Per-condition means at $N = 20$.

Table 7: Per-condition mean α at $N = 20$ fresh seeds.

condition	mean α	std	sem
$- - - -$ baseline	3.008	0.090	0.020
$-C - -$ (C off)	3.464	0.097	0.022
$- - P -$ (P off)	2.790	0.086	0.019
$-CP-$ (both off)	3.628	0.161	0.036

Main effects at $N = 20$. C main effect = +0.456 (turning C off raises α); P main effect = −0.218 (turning P off lowers α).

Interaction estimate. Direct calculation from means:

$$\Delta_{CP} = \frac{(3.628 + 3.008) - (3.464 + 2.790)}{2} = +0.191.$$

Bootstrap on the 20-seed sample (2000 resamples):

- bootstrap mean $\Delta_{CP} = +0.190$;
- 95% bootstrap CI $[+0.143, +0.239]$;
- 0/2000 bootstrap resamples were at or below zero, reported as 0.0000;
- 0/2000 bootstrap resamples were below the preregistered +0.10 floor, reported as 0.0000.

Per-seed paired distribution. 19/20 seeds give a positive paired-interaction estimate (range +0.055 to +0.322); a single seed gives −0.009. No seed gives a strongly negative interaction.

6.4 Reading and disclosure

The 95% CI is entirely above the preregistered $+0.10$ **threshold** on a fresh-seed sample. 0/2000 bootstrap resamples were at or below zero, reported as 0.0000; 0/2000 bootstrap resamples were below the preregistered +0.10 floor, reported as 0.0000.

Architectural reading (substrate witness). C creates churn in *which* vertices are uncrossed (frame rotation churns the uncrossed pool). P promotes the high-pressure subset of the uncrossed pool to mini-emitters. The product is a non-additive novel-event-generation pathway: with both on, the uncrossed pool churns and P amplifies new vertices entering the high-pressure region; with either off, the measured interaction collapses. The interaction +0.19 is comparable in magnitude to the P main effect -0.22, so C and P are *strongly coupled* cascade-state stabilisers on this substrate, not nearly-orthogonal ones. This reverses an architectural claim from the original 3-seed validation that held C and P approximately orthogonal.

Disclosure: $N = 20$ **ordering.** The $N = 20$ deep-dive was conducted *after* the original $N = 3$ failure (2026-04-29 vs 2026-04-20). The seed range 32000–32019 was selected to be non-overlapping with the original 30000s seeds. Two strengthening builds we have not delivered: (i) a second independent $N = 20$ run at a different seed range (e.g. 33000–33019), and (ii) an $N = 50$ characterisation of the per-seed sample distribution. Both are recorded as open builds in §9.

What this stress test does *not* establish.

- It does not establish a Lyapunov function on the reduced flow.
- It does not establish that the substrate is uniquely selected by $C \times P$ coupling among regular 4-polytopes.
- It does not establish an η-trajectory derivation; η is treated as a condition-dependent constant in this paper.

The stress test is what its name says: a high-replication factorial test of one preregistered interaction prediction, on a fresh-seed sample, with bootstrap confidence intervals. The architectural reading is a *description* of what C and P do on the substrate, not a theorem about why they do it.

6.5 Methodological contribution

We document, as a methodological contribution to preregistration practice on this cascade-ablation matrix specifically: in this matrix, P4 ($C \times P$) required $N = 20$ fresh seeds for reliable detection at the preregistered threshold; P3 ($D \times C$) closed at $N = 5$. The original 3-seed preregistered validation gave estimates consistent with underpowered detection on both interaction tests; both close at higher N without threshold modification. The general lesson: when preregistering an interaction effect on a system with unknown per-seed variance, budget the seed count from a power-analysis assumption that the per-seed std could be as large as the interaction effect itself. Future preregistrations on similar high-variance ablation matrices should plan for this scale.

7 Cross-domain selectivity

This section reports three cross-domain witnesses. §7.1 gives the chess pattern-recognition lift. §7.2 gives the conversation neutrality result that confirms the lift is selective. §7.3 gives the HCP brain-graph H_4-transitive null. For each domain we report $A_d = \mathrm{Acc}_{\mathrm{sub},d} - \mathrm{Acc}_{\mathrm{raw},d}$ or, in the HCP case, $Z_m = (m_{\mathrm{ARIA}} - \mu_{\mathrm{HCP}})/\sigma_{\mathrm{HCP}}$. Numbers are lifted verbatim from `docs/brain_mapping/CROSS_DOMAIN_RESULTS.md`.

7.1 Chess pattern recognition (P9–P13)

Setup. 32 chess positions across 4 categories (tactical, positional, endgame, opening). Per-position 8-dimensional V2 features (material balance, king safety, pawn structure, centre control, piece activity, mobility, threat density, defensive structure), normalised by per-feature L^2 norms. Substrate routing: features injected as pressure into the S^7 observer frames; substrate run forward by n_{ticks}; resulting vertex pattern used as classifier feature vector. Classifier: 1-nearest-neighbour on cosine similarity, validated by 5-fold CV or leave-one-out (LOO).

Critical methodological detail. Between successive depth measurements the substrate is reset to canonical state via `mon.homeostatic_reset(level=1.0)`. Without this, pressure-field state drifts toward equilibrium across ~ 5 evaluations and classification structure collapses to raw-feature baseline.

Table 8: Chess substrate-routed depth sweep with state reset between measurements.

n_{ticks}	accuracy
5	53.1%
15	65.6%
25	**93.8**% (← peak)
40	84.4%
60	84.4%
100	78.1%

Table 9: Chess preregistered tests (with reset, $n = 25$ canonical depth).

ID	Test	Threshold	Observed	Verdict
P9	5-fold CV (seeds 30200–30204)	$\geq 70\%$	83.1%	✓
P10	null perm. mapping (chess, 15 perms)†	$\geq 50\%$	65.4%	✓
P11	random-label baseline (20 trials)	$\in [15\%, 35\%]$	23.4%	✓
P12	goldilocks peak depth	$\in \{15, 25, 40, 60\}$	$n = 25$	✓
P13	substrate lift, LOO refinement (with reset)‡	$\geq +15$pp	+**40.6**pp (LOO)	✓

† The 2026-04-18 preregistration combined the null-mapping prediction across both domains ("$\geq$ 50% on chess and conversation"). We split it for table clarity into P10 (chess null) and P16 (conversation null); both pass. The 2026-04-18 preregistration specified 20 random label permutations for the null-mapping bound; the 2026-04-29 validation run used 15 permutations (`run_preregistered_validation.py`; the $\geq$ 50% threshold is unchanged). We report this as a disclosed-protocol deviation, not a threshold change; the observed 65.4% at 15 perms sits well above the 50% floor and the result is robust to perm count in this range. Verification at the preregistered 20-perm setting is an open build (§9).

‡ The 2026-04-18 preregistration P13 specified the chess substrate-lift estimator as 5-fold CV at threshold $\geq$ +15pp. The 2026-04-29 validation tightened the estimator to LOO with state reset; we report the LOO finding (+40.6pp) above as a disclosed estimator/protocol refinement at the unchanged +15pp threshold, not a preregistration revision.

Reading. Substrate routing amplifies chess-position 4-category classification from raw 53.1% (just above 25% chance) to substrate-routed 93.8% at canonical depth $n = 25$. This is a +40.6pp lift on the LOO refinement; on the preregistered 5-fold CV estimator the substrate-routed accuracy is 83.1% (P9), itself well above any reasonable raw-features baseline. The original 2026-04-20 validation reported the LOO lift at +3.1pp, a state-drift artefact closed by the reset protocol (§2).

Permutation null decomposition. The null permutation mapping (P10) randomises the feature→frame assignment, so each feature is routed to a different S^7 frame than canonical.

The substrate retains 65.4% classification accuracy under random permutation — well above the 25% chance level for 4 categories. We read this as a substrate-witness decomposition: a 65.4 percentage-point accuracy floor persists under the architecture-only permutation null (it survives random feature→frame reassignment; the architecture is acting on whatever input lands in the frames), with the remaining $\sim$ 17pp accruing to canonical alignment. We do not claim this decomposition is unique; it is a description of the observed accuracy stack.

7.2 Conversation neutrality (P14–P16)

Setup. 64 utterances across 8 dialogue-act categories. 8-dimensional injection-row features per utterance. Identical substrate routing pipeline to chess.

Table 10: Conversation preregistered tests.

ID	Test	Threshold	Observed	Verdict
P14	raw 5-fold CV (seeds 30220–30224)	$\geq 75\%$	87.5%	✓
P15	substrate lift	$\lvert\cdot\rvert < 10$pp	-4.4pp	✓
P16	null perm. mapping (15 perms)	$\geq 50\%$	70.6%	✓

Reading. Conversation raw features at 87.5% are already strongly discriminative (cf. chess raw $\sim$ 53%); the substrate lift is -4.4pp, well within the preregistered neutrality band $\lvert\cdot\rvert < 10$pp. The substrate is approximately neutral on conversation.

Selective amplifier signature. The pair (chess +40.6pp lift; conversation -4.4pp lift) is consistent with the selective-amplifier behaviour preregistered in 2026-04-18: in these two tasks, the architecture amplifies when raw features are ambiguous (chess raw $\sim$ 53%) and is approximately neutral when raw features are already saturated (conversation raw 87.5%). We do not claim this generalises to all classification tasks; cross-domain transfer to additional ambiguous-feature benchmarks is an open build (§9).

7.3 HCP brain-graph H_4-transitive null (P17–P18)

Setup. Reference cohort: Human Connectome Project (HCP) $n = 1003$ subjects [Van Essen et al., 2013]; preregistered tests on $n = 100$ subjects for computational tractability, with full-cohort $n = 1003$ descriptive statistics also reported. ICA-50 group-averaged connectivity matrix; thresholded at the same density as ARIA's vertex graph ($\rho = 0.101$). Compare degree distribution and higher-order graph statistics. ARIA reference: $G_{V_{600}}$ with $L_{V_{600}}$. By H_4 transitivity (§3.2) every vertex has identical local structure $\Rightarrow$ uniform degree 12 $\Rightarrow$ degree std $= 0$ as a theorem.

Table 11: HCP comparison: preregistered $n = 100$ test plus full-cohort $n = 1003$ descriptive statistics.

Metric	ARIA	HCP $n = 1003$ mean (sd)	σ from HCP
Degree std (preregistered, $n = 100$ subset)	0.000	3.388 (> 2.0)	—
Degree std (descriptive, $n = 1003$)	0.000	3.28 ± 0.28	-11.58σ
Participation ratio (descriptive)	68.54	19.72 ± 0.61	$+79.78\sigma$
Clustering coefficient (descriptive)[♭]	0.455	0.220	$+6.80\sigma$

[♭] The HCP across-subject standard deviation for the clustering coefficient is not separately reported in `CROSS_DOMAIN_RESULTS.md`; the $+6.80\sigma$ value is sourced from the same descriptive analysis as the other rows. Inferred from the reported gap and σ, the implicit HCP sd is $\approx 0.235/6.80 \approx 0.035$. We carry the σ-distance forward as reported and flag the missing explicit sd here.

- P17 (ARIA degree std, theorem): predicted = 0, observed 0.0000, ✓.
- P18 (HCP ICA-50 degree std, $n = 100$ density-matched): predicted > 2.0, observed 3.388, ✓. Zero of 1003 HCP subjects have degree std below 2.0.

Reading (substrate witness). ARIA's H_4-transitive structure is a deterministic group-theoretic null reference for cortical functional connectivity. Real cortex breaks the symmetry through hub-spoke functional specialisation; the σ-distances quantify the magnitude of biological symmetry-breaking with no fitted parameters. The σ-distances (-11.58σ on degree homogeneity, $+79.78\sigma$ on participation ratio, $+6.80\sigma$ on clustering coefficient) are large on the ICA-50 pipeline at the density-matched threshold $\rho = 0.101$; cross-parcellation replication (Schaefer, Glasser) remains an open build.

Participation-ratio comparability. ARIA's vertex graph has 120 nodes; the HCP ICA-50 connectivity matrix has 50 nodes. The participation-ratio statistic $\mathrm{PR}(G) = (\sum_i d_i)^2 / \sum_i d_i^2$ is node-count-dependent — its theoretical maximum is the node count of the graph. We report the raw PR values (ARIA = 68.54 on a 120-node graph; HCP = 19.72 on a 50-node graph) and the σ-distance against the HCP across-subject distribution, but the $+79.78\sigma$ value reflects both the architectural difference and the differing node counts. A node-count-normalised statistic $\mathrm{PR}/|V|$ gives ARIA = 0.571 vs HCP = 0.394, a smaller absolute gap; we keep the raw-PR comparison as headline because the HCP subject distribution and the across-subject σ are computed in the same units, but flag the node-count caveat here.

What we do not claim.

- We do not claim cortex has "drifted from an ideal polytope"; the substrate is a useful a-priori null whose deviation from real cortex is precisely measurable.
- We do not claim parcellation invariance: the σ-distances are reported on ICA-50; alternative parcellations (Schaefer, Glasser) would give different per-metric numbers but, on the basis of the qualitative pattern that cortex is hub-concentrated relative to ARIA's transitive null, we expect them to preserve the signs. Verification across parcellations is an open build (§9).

7.4 Cross-domain summary as a selective amplifier + H_4-transitive null

Table 12: Cross-domain summary on a single substrate.

Task	Raw	Substrate	Null perm.	Geom. floor	Sem. alignment
Chess (LOO, $n = 25$, w/ reset)	53.1%	93.8%	—	—	+40.6pp lift
Chess (5-fold CV)	—	83.1%	65.4%	65.4%	+17.7pp
Conversation (5-fold CV)	87.5%	83.1%	70.6%	70.6%	+12.5pp (substrate vs null)

The geometric content ($\approx$ 65–71% across the two domains) is the architecture-invariant null floor. The semantic content (12–18pp) is the domain-specific contribution. On HCP, σ-distances against the biological cohort are $(-11.58, +79.78, +6.80)$ on (degree std, participation ratio, clustering coefficient).

Headline cross-domain reading. The substrate is *selectively* amplifying (not unconditionally), and it is an H_4-transitive deterministic null on connectivity (not a fitted model). Both readings sit inside the substrate-witness scope.

8 Discussion

This section reads the substrate-witness result against existing theories of consciousness, identifies what is novel here that is not a re-statement of an earlier theory, and proposes a non-load-bearing ACT bridge to the companion adaptive-closure-transport preprint [Smart, 2026a]. We do not claim a selection theorem, we do not claim a Lyapunov derivation, and we do not claim the recurrent layer "is" consciousness.

8.1 What is novel in this work

Three things are claimed novel as a substrate witness:

1. **A geometry-fixed substrate that is consistent with real-cortex EEG signatures without fitted shape parameters on neural data.** Once the 600-cell is chosen as the substrate, its graph (120 vertices, uniform degree 12 on the canonical short-edge nearest-neighbour graph) and the response operator $C_\varphi = L_{V_{600}} + \varphi^{-2} I$ are fixed by the construction (no shape parameter is tuned to neural data); cascade-α matches Sleep-EDFx within preregistered tolerance with pairwise CI overlap on three reference ranges; six drug/sleep signatures pass on a single deterministic recurrent-layer trajectory at seed 42, against literature-derived thresholds. We are not aware of a prior geometric substrate that has been tested against this many preregistered cortical correspondences from a graph fixed by the construction with no neural-data-fitted shape parameters; we cannot rule out that such a model exists.

2. **The strong-coupling architectural finding.** C and P are strongly coupled cascade-state stabilisers, not nearly-orthogonal ones. The $C \times P$ interaction (+0.190, 95% CI $[+0.143, +0.239]$ at $N = 20$) is comparable in magnitude to the P main effect (-0.218). This was hidden by underpowered ablation and emerged only at $N \geq 20$ — a substantive correction to the architectural reading from the original 3-seed validation.

3. **The 18/18 preregistered correspondences with no threshold modification.** Every prediction in the preregistered set passes at the preregistered thresholds. The two interaction tests (P3, P4) required $N \geq 5$ and $N \geq 20$ respectively, and one test (P13) required the documented state-reset protocol. We report this transparently as methodology refinement, not as threshold change.

8.2 Comparison to existing theories of consciousness

vs IIT. [Tononi, 2008, Balduzzi and Tononi, 2008] ARIA produces IIT-direction-correct Φ collapse on propofol (0.33× wake). The Φ proxy (§4) is designed to be small under H_4-equivariant dynamics and to increase when dynamics produce cross-class asymmetries. ARIA does not implement the full IIT axioms (cause-effect repertoires, exclusion postulate, integration-over-partitions); it reproduces an observable consequence on the propofol–wake state contrast. This is a consistency-of-direction result, not a discharge of IIT.

vs Global Workspace Theory. [Baars, 1988, Dehaene, 2014] The S^7 context-rotation mechanism (§4) is functionally analogous to a workspace with rotating attentional selection; the active observer frame plays the role of a temporary in-workspace subset of features. ARIA does not commit to the GWT broadcast/access distinction at the architectural level; the analogy is descriptive.

vs Predictive Processing. [Friston, 2010, Clark, 2013] ARIA does not implement prediction-error minimisation or hierarchical generative models. The recurrent self-model layer ($\eta = 0.20$) provides top-down modulation of the substrate response by cosine direction alignment with the prior phenomenal snapshot, not by learned prediction errors. Predictive-processing-style refinements (e.g. η as an adaptive learning rate over a prediction-error norm) are an open build, not delivered here.

vs neural mass models. ARIA operates at the architectural-algorithmic level; it does not specify which neural circuits implement context rotation or partial emission. The 600-cell substrate is proposed as an abstract description of the criticality- maintaining structure of cortex, not as a circuit model.

8.3 The non-load-bearing ACT bridge

The companion adaptive-closure-transport preprint [Smart, 2026a] proposes a 4-tuple bridge $(M, L_M, W, R_{\text{hom}})$ — substrate M, response operator L_M, learnable Hebbian-like field W, and a homeostatic regulariser R_{hom}. We propose the dictionary D_{ACT}:

$$D_{\text{ACT}}: (M, L_M, W, R_{\text{hom}}) \longmapsto (V_{600},\ C_\varphi,\ \text{cascade pressure field } W_{\text{p}},\ \texttt{homeostatic\_reset}).$$

This bridge is non-load-bearing for the present paper. It is included as a route-K (alternative-route) reading; the substrate- witness claims (six signatures, 18/18, chess +40.6pp, HCP σ-distances) do not require any of the ACT theorems.

What ACT would have to deliver to make this load-bearing. The companion preprint identifies four open builds, each of which is deferred:

- A Lyapunov function $V(W)$ on the reduced flow whose monotonicity proves selection — not delivered.
- An edge-space decomposition of $\mathbb{R}^{E_M}$ under the Hodge edge Laplacian $L_{\text{edge}} = \delta_2\delta_2^\top + \delta_1^\top\delta_1$ — not delivered.
- A formal $2I$-equivariance audit of the closure operator family — not delivered.
- A full reduced-flow convergence theorem on W-trajectories — not delivered.

Until these are delivered, ARIA is positioned as the empirical *substrate witness* for the family that ACT names; ACT is not the selection-theorem witness for ARIA. The companion kernel document [Smart, 2026b] discusses the same C_φ in a passive-regime witness via the $B \to K^*\mu^+\mu^-$ flavour anomaly [Smart, 2026c]; that line shares operator-level infrastructure with this paper, but does not transfer empirical support for ARIA.

8.4 The strong-coupling reading for cortical architecture

Real cortical criticality is maintained by multiple parallel mechanisms: E/I balance, neuromodulation (acetylcholine, noradrenaline), homeostatic plasticity, gain control. The naive expectation — and the one we held until the $N = 20$ deep-dive — is that these are mostly orthogonal, so losing one removes only its own main effect. The $N = 20$ result reverses this on the substrate: C and P are strongly coupled. Disabling one cascades into losing the synergistic contribution of the other.

This matches clinical observations: anaesthesia (which targets GABAergic transmission) and seizure (which targets E/I balance) produce widespread network-level dysfunction beyond their direct targets — qualitatively consistent with a strong-coupling hypothesis. We position this as *a hypothesis the substrate witness raises*, not as a proof. The bridge from cascade-mechanism interaction on V_{600} to real-cortex pharmacological coupling is a step we do not take in this paper.

8.5 Methodological contributions

Two methodological items are worth recording outside the headline:

1. $N \geq 20$ **for similar high-variance ablation matrices.** Allocation discipline for preregistration: in this cascade-ablation matrix specifically, P4 ($C \times P$) required $N = 20$ for reliable detection at the preregistered threshold. The general rule we draw — when preregistering an interaction effect on a system with unknown per-seed variance, budget for at least this scale — should be tested against other ablation matrices, not taken as universal. The original 3-seed plan was the source of two underpowered-interaction estimates in this work.

2. **State-reset protocol on non-stationary substrates.** ARIA's substrate is a non-stationary dynamical system; the pressure field equilibrates within ~ 5 successive evaluations. Any multi-trial benchmark must specify a state-reset protocol or document the drift. Generalisable lesson: *published cross-domain benchmarks on non-stationary substrates should report an explicit reset/equilibration discipline*, not just seed.

8.6 The substrate as an H_4-transitive connectivity null

The HCP comparison (§7.3) places ARIA as a principled deterministic null reference for cortical functional connectivity. Real cortex breaks the symmetry through hub-spoke functional specialisation; the σ-distances from ARIA quantify the magnitude of biological symmetry-breaking with no fitted parameters.

This is a methodological contribution to comparative connectomics. Stochastic nulls (Erdős–Rényi, configuration model, edge-randomised graphs) compare cortex to a random graph with matched density. ARIA is a different kind of null: a deterministic group-theoretic graph with structure-level statements: degree std $= 0$ by H_4 transitivity, and a fully-determined Laplacian spectrum (§3.2) computed from the constructed graph. Both null kinds are useful; ARIA gives a specific, reproducible, group-theoretic baseline that cortex deviates from in quantifiable σ-units.

8.7 Open questions raised by the substrate witness

- Do the six drug/sleep signatures replicate across 10–20 cross-seed runs of the recurrent layer? (Single-seed disclosure; see §9.)
- Do alternative regular 4-polytopes (24-cell, 120-cell) reproduce comparable signature sets, or is the 600-cell distinguished?
- Does the strong-coupling reading ($C \times P$) survive an independent fresh-seed $N = 20$ replication at a different seed range?
- Does the substrate's amplifier behaviour transfer to other ambiguous-feature classification tasks beyond chess (e.g. visual pattern, audio classification)?
- Does the Sleep-EDFx three-way CI overlap survive on a different EEG cohort (TUH, NHM)?

We list these as open questions raised by the witness, not as gaps in the witness itself.

9 Limitations and hostile-review guard matrix

This section enumerates limitations transparently, organised as a five-move guard matrix following the b-anomaly preprint template [Smart, 2026c]: regime, post-hoc, interpretation, test/claim, state-drift. For each guard we record $G\colon \text{risk} \to (\text{disclosure}, \text{evidence}, \text{strengthening build})$.

9.1 Regime

Single substrate (the 600-cell). We have not tested whether other regular 4-polytopes (24-cell, 120-cell) would produce comparable correspondences. The 600-cell was chosen because its H_4 Coxeter cascade structure aligns with the empirical signatures that motivated this paper, not from an a-priori derivation. *Disclosure:* substrate-witness scope, no uniqueness claim (§1). *Evidence:* empirical correspondences hold on V_{600}. *Strengthening build:* formal ablation against $\{\text{24-cell}, \text{120-cell}\}$ on the same six-signature battery and the eighteen preregistered tests, with thresholds preserved.

Single-seed determinism on the recurrent layer. The v4 six-signature results in §5.1 are reported on a single deterministic trajectory at seed 42. Empirical CIs across 10–20 cross-seed runs would strengthen the per-signature claims beyond the single-trajectory bootstrap of 58 events that gives the WAKE 95% CI $[1.82, 2.86]$. *Disclosure:* explicitly single-seed in §2 and §5.1. *Evidence:* bootstrap CI on the wake 58 events, plus three-way overlap with two independent reference α ranges. *Strengthening build:* 10–20 cross-seed runs of `demo_drug_sleep_v4.py`, with per-signature bootstrap CIs.

Stylised stimulus models on the recurrent layer. The v4 stimulus models for WAKE (AR(1) noise + tonic shell + attention episodes), SLEEP_N3 (slow oscillation + spindles + K-complexes), and PROPOFOL (low-amplitude tonic noise) are biologically motivated but abstract: a single shell anchor for tonic coherence, fixed 40-tick period for slow-wave drive, etc. Real spatial structure of cortical input is much richer. The v4 stimulus models were redesigned after diagnostics on the v3 stimulus models to close v3 stimulus-model artefacts; v4 components are biologically-motivated and not fitted to subject-level measurements, but they are condition-specific design choices iterated to v4. *Disclosure:* §4 explicitly frames v4 as a redesign. *Evidence:* amplitudes and durations match published biological time scales; the v3→v4 diff is captured in `CONSCIOUSNESS_CHAIN_V4_SIGNATURES.md`. *Strengthening build:* replication on stimulus models derived from anatomically-grounded input statistics (e.g. retinotopic, tonotopic).

9.2 Post-hoc

The 600-cell choice is post-hoc justified by empirical observables. While the construction of V_{600} is theorem- level rigorous (H_4 Coxeter group theory), the choice of *this* polytope as the consciousness-substrate instance is motivated by the correspondences observed, not by an a-priori biological argument. *Disclosure:* §1, "substrate witness, not derivation; not a uniqueness claim". *Evidence:* the eighteen preregistered correspondences plus six signatures; the H_4 transitivity theorem ($P17$). *Strengthening build:* comparison with the 24-cell and 120-cell on the same signatures; formal ACT-selection-theorem witness via the bridge of §8.3 (deferred).

Cascade decomposition is one of several possible decompositions of H_4. We use the σ-orbit projector basis because it is machine-precise and biologically informative, but other bases (character-theoretic, Galois-twin) give the same physical predictions through different intermediate variables. *Disclosure:* §3.5 acknowledges non-uniqueness of decomposition. *Evidence:* block-diagonalisation at $< 10^{-15}$ cross-block norm. *Strengthening build:* catalogue and equivalence-prove the admissible decompositions.

φ^{-2} floor not derived. The shifted-Laplacian floor $\varphi^{-2} \approx 0.382$ is a stability clamp making C_φ strictly positive definite (§3.3); it is not derived from a closure functional or symmetry argument. *Disclosure:* §3.3 marks this as a design-level choice; the companion kernel doc [Smart, 2026b] explicitly does not derive it. *Evidence:* the same operator (with the same shift) serves as the basis for the b-anomaly passive-regime witness [Smart, 2026c]. *Strengthening build:* derive the φ^{-2} shift as the unique stability clamp under a named regularity criterion.

9.3 Interpretation

The recurrent layer is a method, not a metaphysics claim. We do not claim the recurrent self-model layer "is" consciousness; we claim quantitative consistency with six published biological signatures on a deterministic trajectory. *Disclosure:* §1, §4 ("method, not metaphysics"). *Evidence:* six signatures vs published thresholds. *Strengthening build:* cross-seed CIs (§9.1); a formal account of which substrate observables map to which phenomenal contents (the bind_phenomenal_field channels) is not delivered.

Φ is an IIT-style direction-of-effect proxy, not a full IIT discharge. *Disclosure:* §4 explicitly. *Evidence:* propofol Φ collapse to $0.33\times$ wake matches IIT direction. *Strengthening* *build:* a `phi_iit_full` implementation following Balduzzi–Tononi 2008 [Balduzzi and Tononi, 2008] on the substrate; not delivered.

HCP σ-distances are descriptive, not normative. We do not claim "cortex has drifted from an ideal polytope"; we quantify the distance between cortex and the deterministic H_4 null. *Disclosure:* §7.3 explicitly. *Evidence:* σ-distances on three independent metrics. *Strengthening build:* cross-parcellation replication (Schaefer, Glasser).

9.4 Test/claim

Two preregistered interaction tests required higher N than originally allocated. P3 closes at $N = 5$; P4 closes only at $N = 20$. We document this transparently as consistent with underpowered interaction estimates on high-per-seed-variance terms, not a threshold change. *Disclosure:* §2, §5, §6. *Evidence:* bootstrap CI fully above the +0.10 floor; per-seed distribution narrow at $N = 20$ (std = 0.089); 19/20 seeds positive. *Strengthening build:* a second $N = 20$ run at a different seed range (e.g. 33000–33019); an $N = 50$ characterisation of the per-seed distribution.

The original 2026-04-20 walks-back are reversed without threshold modification. The reversals (P3, P4, P13) are documented in `docs/brain_mapping/VALIDATION_RESULTS_2026-04-29.md` with the original failure values, the methodology refinement, and the post-refinement values. *Disclosure:* this paper carries those disclosures verbatim. *Evidence:* 2026-04-20 vs 2026-04-29 side- by-side results table. *Strengthening build:* the strengthening builds for P3/P4/P13 above; no further claim is needed.

Bayesian and full-IIT inference not performed. All intervals are frequentist (bootstrap CIs); Φ is the direction-of-effect proxy, not the Balduzzi–Tononi 2008 algorithm. *Disclosure:* this section. *Strengthening build:* Bayesian posterior on Δ_{CP}; full-IIT computation on the substrate's S^3 state space. The latter is computationally heavy and is deferred.

9.5 State-drift / out-of-scope

Single condition-dependent parameter η is not derived as a learned variable. $\eta \in \{0, 0.05, 0.20\}$ across PROPOFOL, SLEEP_N3, and WAKE/RECOVERY is a condition-dependent constant in this paper, not a learned trajectory. A predictive-processing extension where η adapts on an error norm is an open build.

No deuteron / hadron / RH / capstone material is loaded into this paper. The companion programme (cascade-derivation, capstone coalgebra, RH formal) shares operator-level infrastructure but is not load-bearing here. Deliberately out of scope.

Out-of-scope items NOT delivered (correctly).

- Active-regime extension of the chess pattern-recognition test to move-by-move game trajectories (this paper covers static positions only).
- Edge-space decomposition of $\mathbb{R}^{E_M}$ under $2I$-equivariance — open build of the ACT companion paper.
- Lyapunov derivation $V(W)$ from a closure functional $\mathcal{F}$ — open build of the ACT companion paper.
- Selection theorem for V_{600} over alternative regular 4-polytopes — see §9.1.
- Cross-cohort EEG (TUH, NHM, OpenNeuro non-propofol) replication of the six signatures.
- Cross-parcellation replication of the HCP σ-distances (Schaefer, Glasser, etc.).
- Bayesian posterior on the C×P interaction.

- Verification of P10 (chess null mapping) at the preregistered 20-permutation count (the 2026-04-29 validation used 15 permutations; threshold $\geq 50\%$ unchanged; result 65.4% robust in the 15–20 range, but the prereg-exact rerun is open).

9.6 The honest verdict

The result is a substrate witness: a geometry-fixed substrate, with no shape parameters tuned to any neural dataset, is consistent with eighteen preregistered correspondences and six companion drug/sleep EEG signatures, with all original gaps closed by methodology refinement and without modifying any preregistered threshold. We do not claim the substrate *is* consciousness. We do not claim a selection theorem on the ACT bridge. We do not claim uniqueness for V_{600} among regular 4-polytopes. The strengthening builds for these stronger claims are listed above and remain open.

10 Conclusion

The 600-cell regular 4-polytope V_{600} under H_4 Coxeter symmetry, with the shifted-Laplacian response operator $C_\varphi = L_{V_{600}} + \varphi^{-2} I$ ($\varphi = (1+\sqrt{5})/2$), is a geometry-fixed substrate that is consistent with eighteen preregistered neuroscience correspondences plus six companion drug/sleep EEG signatures of conscious vs unconscious states. Once the substrate is chosen, its graph structure (120 vertices, uniform degree 12 on the canonical short-edge nearest-neighbour graph, with the Laplacian spectrum reported in §3.2 as observed) is fixed; only one condition-dependent self-injection coupling $\eta \in \{0, 0.05, 0.20\}$ and one substrate-pinned nonlinearity bounded_topk$(\cdot, k=12)$ at the graph's average degree enter the recurrent layer above the substrate. No shape parameter is tuned to any neural dataset.

Headline tally. On a single deterministic trajectory, six drug/sleep EEG signatures pass against their literature-derived thresholds (Sleep-EDFx CI, OpenNeuro `ds005620`, Brodbeck 2012, Tononi 2008): NREM-N3 phenomenal-intensity variance ratio 0.463× wake; propofol modality-switching 1.83× wake; propofol continuity drop +0.066; propofol integrated-information Φ collapse to 0.33× wake (IIT direction confirmed); recovery deterministically identical to wake under the WAKE stimulus protocol; wake cortical-avalanche power law $\alpha = 2.252$, 95% CI $[1.82, 2.86]$, $R^2 = 0.956$. The wake 95% CI overlaps the real Sleep-EDFx EEG 95% CI ($n = 30$ subjects, $\alpha = 2.51$, CI $[2.50, 2.53]$) and ARIA's prior cascade pipeline CI $[2.73, 3.25]$.

Eighteen preregistered correspondences. All eighteen pass at preregistered thresholds, with two interaction tests requiring $N \geq 5$ and $N = 20$ respectively for reliable detection of high-variance interaction terms, and one cross-domain test requiring the documented `homeostatic_reset` state-reset protocol. No preregistered threshold has been modified. The original 2026-04-20 15/18 tally was a methodology-limited reading, not a content failure; the closure of the three gaps (P3, P4, P13) is documented transparently in `docs/brain_mapping/VALIDATION_RESULTS`

Strong-coupling architectural finding. Two cascade mechanisms — context rotation C and partial emission P — are causally identifiable within the factorial ablation model and exhibit strong synergy: their interaction $\Delta_{CP} = +0.190$ at $N = 20$ (95% bootstrap CI $[+0.143, +0.239]$, 0/2000 resamples at or below zero, reported as 0.0000) is comparable in magnitude to the P main effect -0.218. The original 3-seed estimate (+0.044) is consistent with an underpowered interaction estimate on a high-per-seed-variance term (std = 0.089 at $N = 20$); we contribute $N \approx 20$ as a planning scale for this cascade matrix, recommended as a preregistration-practice consideration for similar high-variance ablation matrices.

Cross-domain selectivity. The substrate exhibits selective amplification on the two tasks tested: chess 4-category position classification on 8-D V2 features lifts +40.6pp on leave-one-out at canonical depth $n = 25$ ticks (raw 53.1% $\rightarrow$ substrate-routed 93.8%, with state reset; preregistered threshold $\geq$ +15pp on 5-fold CV — the LOO finding above is a disclosed estimator/protocol refinement at the same threshold), while conversation utterance classification at raw 87.5% lifts −4.4pp (threshold $|\cdot| < 10$pp) — and as an H_4-transitive deterministic null reference for cortical functional connectivity: on the full-cohort descriptive HCP $n = 1003$ statistics (preregistered test on the $n = 100$ subset), ARIA's H_4-transitive structure is at -11.58σ on degree homogeneity, $+79.78\sigma$ on participation ratio (with the node-count caveat of §7.3), and $+6.80\sigma$ on clustering coefficient.

Substrate-witness scope. This is a substrate witness, not a derivation of consciousness, not a selection theorem on the companion adaptive-closure-transport 4-tuple [Smart, 2026a], and not a uniqueness claim for the 600-cell among regular 4-polytopes. The strengthening builds — cross-seed CIs on the recurrent-layer signatures, alternative-polytope ablations, an independent $N = 20$ C×P replication at a different seed range, cross-parcellation HCP replication, a Lyapunov function on the reduced flow, $2I$-equivariance audit of the closure operator family — are explicitly listed in §9 and remain open.

We are not aware of a prior deterministic geometric architecture tested against this many preregistered cortical correspondences from a graph fixed by the construction with no shape parameters tuned to neural data; we cannot rule out that such prior work exists. The empirical material gathered here is the substrate witness; the broader programme to turn the witness into a selection-theorem-grade claim — including the independent passive-regime witness via the $B \to K^*\mu^+\mu^-$ flavour anomaly [Smart, 2026c] on the same response operator C_φ — is sketched in the companion preprints and remains the natural next step.

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Reproducibility

The complete pipeline (substrate construction, six-signature consciousness chain, $N = 20$ C×P deep-dive, eighteen-prediction preregistered validation, figure regeneration, this paper) is reproducible from the project repository [Smart, 2026d] via the included `reproduce_paper_claims.sh` script. All scripts are deterministic given seeds; the substrate's spectral decomposition is cached at module level. Wallclocks: drug/sleep v4 ~ 30 s; $N = 20$ deep-dive ~ 28 min; preregistered validation ~ 18 min.

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